DEPARTMENT 1 · DEVELOPMENT AND IT · PROFESSION 2

AI integration developer

Profession 2 of 10 in Development and IT.

What the profession means

An AI integration developer connects AI models to the software a business already uses: the website, the CRM, the help desk, the spreadsheet. The work is about data flow, reliability, cost and safety, not about training models.

What professionals do every day

Reads the business process, chooses the model and tool, writes the API calls, handles errors and rate limits, validates outputs, logs decisions and monitors cost.

Why businesses hire them, and what problems they solve

Businesses have data in many systems. An integration makes an AI step part of the process, such as classifying enquiries or drafting replies, without changing the whole system.

- Repetitive reading and sorting of enquiries.
- Summaries and drafts that staff must currently write from scratch.
- Data scattered across tools.

Skills you need

- APIs, JSON and webhooks (Development and IT textbook, Part 1).
- Prompting and structured output (JSON with a schema).
- Error handling, retries and rate limits.
- Security: keys, personal data, logs.
- Cost control: tokens, caching, model choice.

Tools (free and paid)

Tool	What it is for	Free option	Paid option	Official link (not checked live)
OpenAI API	Models for text, classification and tools	Pay per use	Pay per use	platform.openai.com/docs
Anthropic API	Claude models for text and tools	Pay per use	Pay per use	docs.claude.com
Google AI Studio	Try and build with Gemini	Free tier	Pay per use	aistudio.google.com
n8n	Connect apps with AI steps	Self-hosted free	Cloud plans	n8n.io
Postman	Test API calls	Free	Paid teams	www.postman.com

Learning sequence: beginner, intermediate, advanced

Level	What you can do when you finish
Beginner	A script that sends one text to a model and gets JSON back.
Intermediate	A workflow that classifies form submissions and writes them to a sheet.
Advanced	A CRM integration with validation, logging and a human review queue.

Projects by level

- **Beginner:** Enquiry classifier for a clinic.
- **Intermediate:** Draft-reply assistant with approval.
- **Advanced:** Multi-system pipeline with monitoring and cost report.

Client-ready portfolio project

An AI-assisted enquiry pipeline for a fictional business: form to model to CRM, with validation, approval for sends, logging and a cost estimate.

Worked example

Worked example. A form submission is sent to a model with an instruction to return `{"category": "..."}`. Your code checks the category is one of four allowed values; anything else goes to a human. The rule lives in code, not only in the prompt.

Assignment with full solution

ASSIGNMENT

Classify and validate enquiries

Business problem: A clinic receives enquiries by web form. Staff want the enquiry sorted into booking, price or complaint, with the wrong answers caught before they reach anyone.

Steps

1. Write a prompt that returns only JSON with one field, category, and the three allowed values.
2. Call the model (see the example) and parse the JSON.
3. Reject any value outside the allowed list and send that enquiry to a human.
4. Log the input id, category and time. Do not log personal details.
5. Test with ten sample enquiries, including one that is ambiguous.

Expected output: A function that returns one of three categories, rejects others, and logs safely.

SOLUTION (ANSWER KEY)

The prompt asks for JSON only; the code validates it, because models sometimes add text or invent values. Ambiguous enquiries are sent to a human. The log holds an id and the category, never the name or phone number. Test the rejection path as carefully as the success path.

PYTHON

```
ALLOWED = {"booking", "price", "complaint"}
def classify(enquiry, call_model):
    raw = call_model(f"Return only JSON {{\"category\": one of booking|price|complaint}} for:
{enquiry}")
    try:
        category = json.loads(raw)["category"]
    except (ValueError, KeyError):
        return "human_review"
    return category if category in ALLOWED else "human_review"

# offline test with a fake model
import json
assert classify("Can I book Monday?", lambda p: '{"category": "booking"}') == "booking"
assert classify("Refund please", lambda p: "Sure, it's a complaint!") == "human_review"
assert classify("Hi", lambda p: '{"category": "spam"}') == "human_review"
print("integration tests passed")
```

Common mistakes

- Trusting the model's output without validation.
- Logging personal data.
- Sending messages without a review step.
- Ignoring cost: long inputs multiply the bill.

Freelancing opportunities

- Automation and AI integrations for small businesses.
- Maintenance contracts for existing AI features.
- Freelance briefs for 'OpenAI API', 'Claude API', 'AI integration', 'webhook'.

Learning resources

LEARNING RESOURCES (VERIFY BEFORE USE)

- OpenAI API documentation
- Claude documentation
- Google AI for developers

Part of **Master Freelancing: AI and Automation**. Back to the chapter list on skillnavora.in · Open the full textbook. Official links in this chapter were not checked live; confirm before relying on them.